



Faculty of Engineering

Department of Biomedical Engineering

Non-Contact Type Clinical Thermometer Acceptance Test Procedure

PART NUMBER: 01-MIY-013

REV: 1

DATE:16.05.2026





CHANGE RECORDS

FIRST RELEASE

CREATED BY

Name	Title	Date	Sign.
Gülşen Demir	System Engineer	16.05.2026	

CHECKED BY

Name	Title	Date	Sign.
Fadime Taş	Quality Engineer	16.05.2026	

APPROVED BY

Name	Title	Date	Sign.
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REVISIONS

Rev. No	Change Description	Date	QA Approval



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1. SCOPE

The scope of this Acceptance Test Procedure is to formally validate the performance of the non-contact thermometer. The device will be tested against a controlled thermal plate and a commercial reference thermometer to verify its accuracy, operational controls, and functional limits.

2. REFERENCES

01-MIY-002: System Requirement Specification, Contact Body Thermometer
01-MIY-013: Verification-Validation Test Plan for Contact Body Thermometer

3. TEST EQUIPMENT AND SET-UP

3.1 TEST EQUIPMENT

NO	MODEL	DESCRIPTION	CALIBRATION
1	DC Power Supply	DC Power Supply	NO
2	Hot Plate	Temperature controlled hot plate	NO
3	MLX90614	Non- Contact Thermometer	YES
4	Reference Thermometer	Handheld, off-the-shelf commercial infrared thermometer used to compare results.	NO
5	PC	Connected to our thermometer to monitor and log the serial data and temperature readings.	NO

3.2 TEST SET-UP

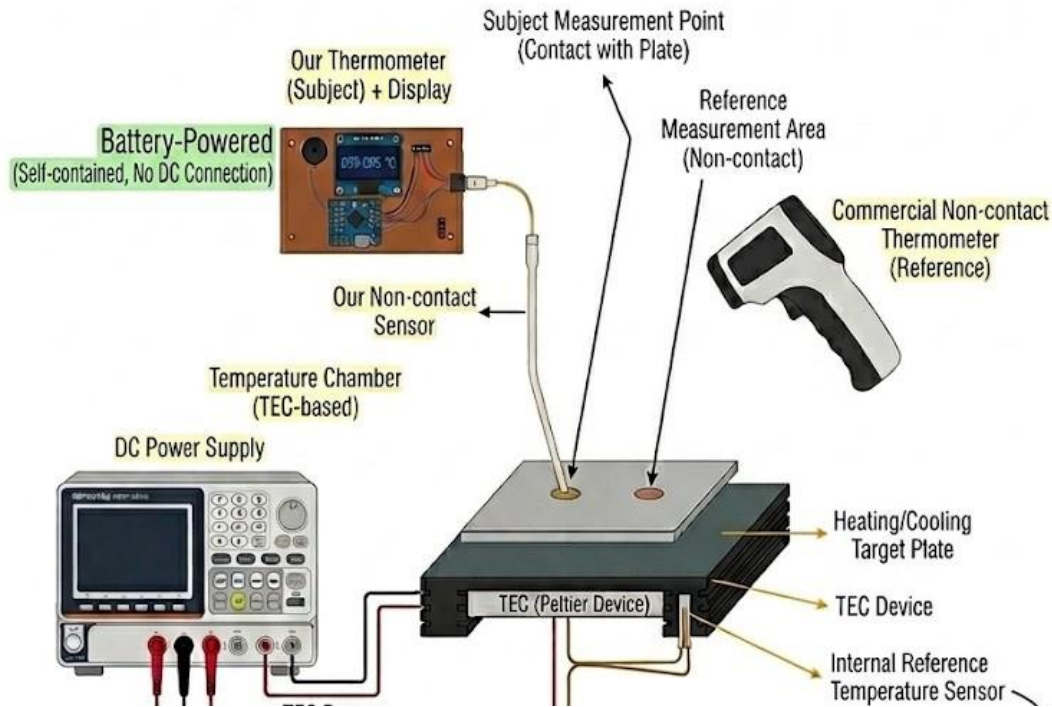


Figure-1: Test set-up for thermometer comparison

4. TEST PROCEDURE

4.1 OPERATING CONTROLS

4.1.1 MODE SWITCH

1. Actuate the physical momentary push-button switch installed on thermometer.



2. Verify that the thermometer must reliably change its operational status.

4.1.2 TURN ON / AUTOMATIC TURN OFF

1. Make the connections as shown on Figure-1.
2. Ensure that the battery is properly connected to the thermometer circuit.
3. Turn the power switch to the ON position.
4. Verify that the OLED display powers on successfully.
5. Position the thermometer toward the target surface at the specified measurement distance.
6. Press the measurement button and verify that the temperature value is displayed correctly.
7. Leave the thermometer inactive after measurement completion.

4.2 ACCURACY

1. Make the connections as shown in Figure-1.
2. Ensure that the power supply connected to the hot plate is initially turned OFF.
3. Place the reference thermometer on the heated plate surface.
4. Turn ON the power supply and gradually heat the plate.
5. Monitor the reference thermometer until the hot plate temperature reaches 37°C, then turn OFF the power supply.
6. Wait until the temperature stabilizes. The plate may be considered stable when the reference thermometer temperature variation is less than 0.1°C for 10 seconds.
7. Turn ON the non-contact infrared thermometer (DUT).
8. Aim the infrared sensor toward the hot plate measurement surface from the defined measurement distance.
9. Measure the hot plate temperature simultaneously using both the reference thermometer and the DUT.
10. Record both measurement values.
11. Verify that the temperature difference between the reference thermometer and the DUT is less than $\pm 0.3^{\circ}\text{C}$.



4.3 RATED OUTPUT RANGE

1. Make the connections as shown in Figure-1.
2. Ensure that the power supply output is initially disabled.
3. Turn ON the power supply and heat the hot plate until the reference thermometer reads 34°C.
4. Turn OFF the power supply and wait for temperature stabilization.
5. Turn ON the infrared thermometer and measure the hot plate temperature.
6. Record the measurements from both the reference thermometer and DUT.
7. Verify that the temperature difference is less than $\pm 0.3^{\circ}\text{C}$.
8. Repeat Steps 1 through 7 by increasing the hot plate temperature to 43°C.
9. Verify that the temperature difference remains less than $\pm 0.3^{\circ}\text{C}$ throughout the operating range.

4.4 UNITS OF MEASURE

1. Initialize the non-contact thermometer by toggling the power switch.
2. Observe the data presentation layout on the 0.96" OLED screen.
3. Verify that the measured temperature value is explicitly accompanied by the metric unit symbol for degrees Celsius: "°C".

4.5 MEASUREMENT TIME

1. Make the connections as shown in Figure-1.
2. Ensure that the hot plate temperature is stabilized at 37°C using the reference thermometer.
3. Keep the infrared thermometer at ambient room temperature for at least 10 minutes before testing.
4. Turn ON the thermometer device.
5. Aim the infrared sensor toward the hot plate surface from the specified measurement distance.
6. Press the measurement button to begin temperature acquisition.
7. Measure and record the elapsed time between pressing the button and displaying the final stable temperature reading.

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4.6 ALARM CONDITIONS

4.6.1. COMPLETION OF MEASUREMENT

1.Completion of measurement is verified at the “Measurement Time” test.

4.6.2. OUTPUT TEMPERATURE OUTSIDE THE RATED OUTPUT RANGE

1. Configure the test bench by ensuring the reference (REF) thermometer is positioned directly on the heated plate surface.
2. Energize the power supply to heat the plate surface. Monitor the REF thermometer until the surface temperature reaches a baseline of 37°C, then adjust the power supply to maintain stability.
3. Power on the non-contact infrared thermometer (Device Under Test - DUT), align the infrared camera hole perpendicular to the plate at the operational distance, and press the measurement button.
4. Verify that the OLED display accurately registers and outputs the standard hot plate temperature.
5. Adjust the power supply (or allow the plate to cool) while observing the REF thermometer. When the plate surface temperature drops and stabilizes at 32°C lock in the state.
6. Align the infrared camera hole with the 32°C plate surface and press the measurement button.
7. Verify that the OLED display reliably prompts the low out-of-bounds indicator reading "**Low**".
8. Energize the power supply again to increase the temperature of the plate surface. Monitor the REF thermometer until the surface temperature reaches 40°C, then stabilize the input.
9. Align the infrared camera hole with the 40°C plate surface and press the measurement button.
10. Verify that the OLED display reliably prompts the high out-of-bounds indicator reading "**High**".

5. TEST REPORT

1. Test results will be recorded on the Acceptance Test Data Sheet in the Attachment-1

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ACCEPTANCE TEST DATA SHEET

Assembly P/N:	001-MIY-087	Date:	14.05.2026
Assembly S/N:	001-MIY-089	Test Operator:	Gülşen Demir
Test Procedure:	Acceptance Test	QA	Fadime Taş

ATP paragraph	Description	Test Limits	Measurement Result	Pass/Fail
4.1	Physical Requirements			
4.1.1	Weight	Inspection	113 gr	Pass
4.1.2	Surface	Demonstration	Pla	Pass
4.2	Functional Requirements			
4.2.1	Operating Controls	Test	Measurements are taken by pressing the buttons; otherwise, it stops taking measurements.	Pass
4.2.2	Power supply	Demonstration	3.7 V rechargeable battery	Pass
4.2.3	Units of Measure	Demonstration	°C	Pass
4.2.4	Rated Output Range	Test	25 °C to 45° C	Pass
4.2.5	Measurement Time	Test	0-1 seconds	Pass
4.2.6	Alarm Conditions	Test	N/A	Fail
4.2.7	Accuracy	Test	±0.3	Pass
4.3	Environmental Conditions			
4.3.1	Operating Temperature	Test	10° C to 40° C	Pass
4.3.2	Storage Temperature	Test	-25 °C to 55 °C	Pass
4.4	Usability			
4.4.1	Display Size	Inspection	0.96 inch	Pass
4.4.2	Measuring Site	Demonstration	Forehead	Pass

N/A: Not Applicable

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INTERNAL AUDIT CHECK LIST*

Auditor: Fadime Taş		Signature: 	
No	Description	Check	CAR#
RESPONSIBILITIES, AUTHORITY			
1	Project Organization Chart is available in Project Plan	+	
2	Job Description is made for Key Personnel in Project Plan	+	
3	Review & Approval Responsibilities are specified Project Plan	+	
DOCUMENT & DATA CONTROL			
4	Document Numbering System is established and applied for documentation	+	
5	Necessary documents are prepared, review & approved	+	
6	There is a document List	+	
PURCHASING			
7	BOM lists have sufficient information for purchasing	+	
PRODUCT IDENTIFICATION & TRACEABILITY			
8	Each component and assembly have a unique Part Number and has a serial number when necessary	+	
PROCESS CONTROL			
9	Technical & Engineering Specifications are available	+	
10	Applicable related Standards are referenced on documents	+	
11	Work Break Down Structure is available in Project Plan	+	
INCOMING INSPECTION & TEST			
12	In-house manufactured materials and Incoming materials are inspected	+	
IN-PROCESS INSPECTION & TEST			
13	Quality Control Inspections & Test Plans are available	+	
14	Necessary test equipment is available in laboratories	+	
FINAL INSPECTION & TEST			
15	Test Procedures are sufficient to perform the tests	+	
16	Test results and inspection results are documented		
CALIBRATION			
17	Calibration Certificates for equipment & devices are available	+	
CONTROL OF NON-CONFORMING PRODUCT			
18	Corrective-Preventive Action are taken for non-conforming materials	+	
CORRECTIVE & PREVENTIVE ACTIONS			
19	Corrective -Preventive Actions are taken for processes	+	
20	Non-conformity Follow-Up is implemented for Corrective Actions	+	
21	Corrective-Preventive Actions are closed appropriately	+	
22	Corr-Prev. Actions are reviewed during Review Meetings	+	
INTERNAL AUDIT			
23	Internal Audit is carried out and documented	+	

*This check list is extracted from TMMOB MAKİNA MÜHENDİSLERİ ODASI "Internal Audit Check List"

-	The audit shall be performed by Quality Assurance Engineer
-	"Check" column shall be checked for acceptable processes
-	A "Corrective Action Report" shall be initiated for non-conformances



NON-CONFORMING MATERIAL REPORT (NCMR)

NCMR No: 1	Date: 14.05.24	Initiated by: PE	
Part No: 1	Description: Housing Assembly / Enclosure		
Serial No:	Process: Identification, Correction, Documentation		
Non-Conformance Description: During the KIT Placement assembly step, it was identified that the 3D-printed housing components were oversized. The internal cavity and mounting dimensions did not match the required specifications to securely hold the PCB and associated electronic hardware. Inspection Findings: Improper Dimensions: The internal tolerances of the housing were too large, preventing the main board, display, and buttons from aligning properly with the exterior cutouts. The components could not be securely fixed within the enclosure.			
Corrective Action: The 3D CAD model of the housing was revised to decrease the internal volume and adjust the dimensional tolerances for a precise fit. A new housing unit was fabricated using the updated model, allowing the assembly to be successfully completed.			
Disposition: Rework or scrap and replace	MRB Approval	PE/PCE Approval for closure	Date: 16.05.2026
Rework ☒			
Repair* ☐		QAE Approval for closure:	Close Date: 16.05.2026
Use As Is* ☐		Fadime Taş	



CORRECTIVE ACTION REPORT (CAR)

CAR Number: 1		Process:	
Reported by: Quality Assurance Engineer		18.05.2026	
Non-conformance Description: Hardware Defect: The upper half of the OLED display is completely non-functional, restricting the available visual area for temperature readings and user interface elements..			
Responsibility: Software Engineer			
Root Cause: Root cause analysis identified physical damage or a hardware defect in the display module, which caused the top 32 pixel rows to fail.			
Corrective Action Taken: To ensure continuous operation without immediate hardware replacement, the device's firmware was completely optimized. All graphical animations, temperature outputs, and status texts were strictly constrained to the functional bottom half of the screen, maintaining full readability and device usability.			
Completion Date: 15.05.2026		Responsible Signature: 	
QAE Disposition:	☒ Acceptable	☐ Not Acceptable	
QAE Approval: 	Closure Date: 15.05.2026		



CORRECTIVE ACTION REPORT (CAR)

CAR Number: 2		Process:	
Reported by: Program Engineer		16.05.2026	
Non-conformance Description: Solder: During technical inspection, disconnected solder joints were detected on the component pins of the sensor and microcontroller feedback loops, causing inaccurate temperature readings.			
Responsibility: Production Engineer			
Root Cause: Solder joint degradation over time caused increased resistance on the feedback loop component pins, leading to software/sensor communication errors.			
Corrective Action Taken: Precision manual soldering rework was performed to repair all cold solder joints and bridges on the sensor and microcontroller units.			
Completion Date: 17.05.2026		Responsible Signature: 	
QAE Disposition:	☒ Acceptable	☐ Not Acceptable	
QAE Approval:		Closure Date: 17.05.2026	



CORRECTIVE ACTION REPORT (CAR)

CAR Number: 3	Process:
Reported by: Program Engineer	16.05.2026
Non-conformance Description: Solder: During technical inspection, and for better fit for inner space of 3d model we decided to change battery size and battery itself from 1000mAh to 120mAh. Brand name and voltage is same.	
Responsibility: Production Engineer	
Root Cause: It would not fit into housing properly	
Corrective Action Taken: New Battery equipped for thermometer	
Completion Date: 17.05.2026	Responsible Signature: 
QAE Disposition:	☒ Acceptable ☐ Not Acceptable
QAE Approval: 	Closure Date: 17.05.2026